

Math 166
Integration strategies

Names: _____

Section Number: _____

Goal. Evaluate the following integrals by any method

1. $\int \frac{x^2}{\sqrt{x-5}} dx$

2. $\int \cos(\sqrt{x}) dx.$

3. $\int \frac{1}{e^{3x} - e^x} dx$

4. Evaluate $\int \cos(2x)\sqrt{1 + \sin^2(2x)} dx$.

Hint: You may use that $\int \sec^3(x)dx = \frac{1}{2}(\sec(x)\tan(x) + \ln|\sec(x) + \tan(x)|) + C$